



NATURAL LANGUAGE
PROCESSING



PROJECT

GRAMMATICAL STRUCTURE ANALYSIS

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POS Tagging and Dependency Parsing for Grammatical Structure Analysis using NLP

Objective

To analyze the grammatical structure of text using Part-of-Speech tagging and dependency parsing with NLP libraries.

POS tagging was performed using spaCy's pre-trained English model to identify grammatical categories of words. Dependency parsing was applied to analyze syntactic relationships between tokens. The dependency tree visualization helped in understanding subject–predicate relationships and sentence structure.

Tools & Libraries

- Python
- spaCy
- displaCy (visualization)

Methodology

- Text preprocessing
- Tokenization
- POS tagging
- Dependency parsing
- Visualization

Loading the Text Corpus

```
text = """The Impact of Artificial Intelligence on Modern Society is Artificial Int  
At its core, artificial intelligence refers to the ability of machines to simulate
```

One of the most impactful applications of AI is in healthcare. AI-powered systems a
"""

Loading spaCy Model

```
# Importing Required Libraries
import spacy
nlp = spacy.load('en_core_web_sm')
```

```
doc = nlp(text)
```

```
doc
```

The Impact of Artificial Intelligence on Modern Society is Artificial Intelligence (AI) has emerged as one of the most transformative technologies of the twenty-first century. From simple rule-based systems to complex deep learning architectures, AI has evolved rapidly and is now deeply integrated into many aspects of daily life. Its influence can be seen across industries such as healthcare, education, finance, transportation, and entertainment. As AI systems continue to grow in capability and adoption, they are reshaping how humans work, communicate, and make decisions.

At its core, artificial intelligence refers to the ability of machines to simulate human intelligence. This includes learning from data, recognizing patterns, understanding language, and making decisions with minimal human intervention. Early AI systems were limited in scope and required explicit programming. However, modern AI relies heavily on machine learning and deep learning techniques, enabling systems to improve automatically through experience. The availability of large datasets, increased computational power, and advanced algorithms has accelerated this progress significantly.

One of the most impactful applications of AI is in healthcare. AI-powered systems are now used to assist doctors in diagnosing diseases, predicting patient outcomes, and personalizing treatment plans. For example, machine learning models can analyze medical images such as X-rays and MRIs with high accuracy, often matching or exceeding human performance. AI also plays a crucial role in drug discovery by analyzing chemical compounds and predicting their effectiveness, reducing both time and cost. Despite these benefits, concerns remain regarding data privacy, bias in medical datasets, and the need for human oversight in critical healthcare decisions.

POS Tagging Implementation

```
for token in doc:
    print(f"{token.text:<15} {token.pos_:<10} {token.tag_:<10} {token.dep_}")
```

The	DET	DT	det
Impact	PROPN	NNP	nsubj
of	ADP	IN	prep
Artificial	PROPN	NNP	compound
Intelligence	PROPN	NNP	pobj
on	ADP	IN	prep
Modern	PROPN	NNP	compound
Society	PROPN	NNP	pobj
is	AUX	VBZ	ROOT
Artificial	PROPN	NNP	compound
Intelligence	PROPN	NNP	attr
(	PUNCT	-LRB-	punct
AI	PROPN	NNP	appos
)	PUNCT	-RRB-	punct
has	AUX	VBZ	aux
emerged	VERB	VRB	conj
as	ADP	IN	prep
one	NUM	CD	pobj
of	ADP	IN	prep
the	DET	DT	det
most	ADV	RBS	advmod
transformative	ADJ	JJ	amod
technologies	NOUN	NNS	pobj
of	ADP	IN	prep
the	DET	DT	det
twenty	NUM	CD	advmod
-	PUNCT	HYPH	punct
first	ADJ	JJ	amod
century	NOUN	NN	pobj
.	PUNCT	.	punct
From	ADP	IN	prep
simple	ADJ	JJ	amod
rule	NOUN	NN	npadvmod
-	PUNCT	HYPH	punct
based	VERB	VRB	amod
systems	NOUN	NNS	pobj
to	PART	TO	prep
complex	ADJ	JJ	amod
deep	ADJ	JJ	amod
learning	NOUN	NN	pobj
architectures	VERB	VBZ	pobj
,	PUNCT	,	punct
AI	PROPN	NNP	nsubj
has	AUX	VBZ	aux
evolved	VERB	VRB	ROOT
rapidly	ADV	RB	advmod
and	CCONJ	CC	cc
is	AUX	VBZ	auxpass
now	ADV	RB	advmod
deeply	ADV	RB	advmod
integrated	VERB	VRB	conj
into	ADP	IN	prep
many	ADJ	JJ	amod
aspects	NOUN	NNS	pobj
of	ADP	IN	prep
daily	ADJ	JJ	amod

life	NOUN	NN	pobj
.	PUNCT	.	punct
Its	PRON	PRP\$	poss
influence	NOUN	NN	nsubjpass
can	AUX	MD	aux
be	AUX	VB	auxpass
seen	VERB	VCN	ROOT
across	ADP	IN	prep
industries	NOUN	NNS	pobj
such	ADJ	JJ	amod
as	ADP	IN	prep
healthcare	NOUN	NN	pobj
,	PUNCT	,	punct
education	NOUN	NN	conj
,	PUNCT	,	punct
finance	NOUN	NN	conj
,	PUNCT	,	punct
transportation	NOUN	NN	conj
,	PUNCT	,	punct
and	CCONJ	CC	cc
entertainment	NOUN	NN	conj
.	PUNCT	.	punct
As	SCONJ	IN	mark
AI	PROPN	NNP	compound
systems	NOUN	NNS	nsubj
continue	VERB	VBP	advcl
to	PART	TO	aux
grow	VERB	VB	xcomp
in	ADP	IN	prep
capability	NOUN	NN	pobj
and	CCONJ	CC	cc
adoption	NOUN	NN	conj
,	PUNCT	,	punct
they	PRON	PRP	nsubj
are	AUX	VBP	aux
reshaping	VERB	VBG	ROOT
how	SCONJ	WRB	advmod
humans	NOUN	NNS	nsubj
work	VERB	VBP	ccomp
,	PUNCT	,	punct
communicate	VERB	VBP	conj
,	PUNCT	,	punct
and	CCONJ	CC	cc
make	VERB	VB	conj
decisions	NOUN	NNS	dobj
.	PUNCT	.	punct

	SPACE	_SP	dep
At	ADP	IN	prep
its	PRON	PRP\$	poss
core	NOUN	NN	pobj
,	PUNCT	,	punct
artificial	ADJ	JJ	amod
intelligence	NOUN	NN	nsubj
refers	VERB	VBZ	ROOT

to	ADP	IN	prep
the	DET	DT	det
ability	NOUN	NN	pobj
of	ADP	IN	prep
machines	NOUN	NNS	pobj
to	PART	TO	aux
simulate	VERB	VB	acl
human	ADJ	JJ	amod
intelligence	NOUN	NN	dobj
.	PUNCT	.	punct
This	PRON	DT	nsubj
includes	VERB	VBZ	ROOT
learning	VERB	VBG	xcomp
from	ADP	IN	prep
data	NOUN	NNS	pobj
,	PUNCT	,	punct
recognizing	VERB	VBG	advcl
patterns	NOUN	NNS	dobj
,	PUNCT	,	punct
understanding	VERB	VBG	acl
language	NOUN	NN	dobj
,	PUNCT	,	punct
and	CCONJ	CC	cc
making	VERB	VBG	conj
decisions	NOUN	NNS	dobj
with	ADP	IN	prep
minimal	ADJ	JJ	amod
human	ADJ	JJ	amod
intervention	NOUN	NN	pobj
.	PUNCT	.	punct
Early	ADJ	JJ	amod
AI	PROPN	NNP	compound
systems	NOUN	NNS	nsubjpass
were	AUX	VBD	auxpass
limited	VERB	VCN	ROOT
in	ADP	IN	prep
scope	NOUN	NN	pobj
and	CCONJ	CC	cc
required	VERB	VBD	conj
explicit	ADJ	JJ	amod
programming	NOUN	NN	dobj
.	PUNCT	.	punct
However	ADV	RB	advmod
,	PUNCT	,	punct
modern	ADJ	JJ	amod
AI	PROPN	NNP	nsubj
relies	VERB	VBZ	ROOT
heavily	ADV	RB	advmod
on	ADP	IN	prep
machine	NOUN	NN	compound
learning	NOUN	NN	pobj
and	CCONJ	CC	cc
deep	ADJ	JJ	amod
learning	NOUN	NN	compound
techniques	NOUN	NNS	conj
,	PUNCT	,	punct

enabling	VERB	VBG	advcl
systems	NOUN	NNS	dobj
to	PART	TO	aux
improve	VERB	VB	xcomp
automatically	ADV	RB	advmod
through	ADP	IN	prep
experience	NOUN	NN	pobj
.	PUNCT	.	punct
The	DET	DT	det
availability	NOUN	NN	nsubj
of	ADP	IN	prep
large	ADJ	JJ	amod
datasets	NOUN	NNS	pobj
,	PUNCT	,	punct
increased	VERB	VBD	ROOT
computational	ADJ	JJ	amod
power	NOUN	NN	dobj
,	PUNCT	,	punct
and	CCONJ	CC	cc
advanced	ADJ	JJ	amod
algorithms	NOUN	NNS	nsubj
has	AUX	VBZ	aux
accelerated	VERB	VRB	conj
this	DET	DT	det
progress	NOUN	NN	dobj
significantly	ADV	RB	advmod
.	PUNCT	.	punct

	SPACE	_SP	dep
One	NUM	CD	nsubj
of	ADP	IN	prep
the	DET	DT	det
most	ADV	RBS	advmod
impactful	ADJ	JJ	amod
applications	NOUN	NNS	pobj
of	ADP	IN	prep
AI	PROPN	NNP	pobj
is	AUX	VBZ	ROOT
in	ADP	IN	prep
healthcare	NOUN	NN	pobj
.	PUNCT	.	punct
AI	PROPN	NNP	npadvmod
-	PUNCT	HYPH	punct
powered	VERB	VRB	amod
systems	NOUN	NNS	nsubjpass
are	AUX	VBP	auxpass
now	ADV	RB	advmod
used	VERB	VRB	ROOT
to	PART	TO	aux
assist	VERB	VB	xcomp
doctors	NOUN	NNS	dobj
in	ADP	IN	prep
diagnosing	VERB	VBG	pcomp
diseases	NOUN	NNS	dobj
,	PUNCT	,	punct

predicting	VERB	VBG	advcl
patient	ADJ	JJ	amod
outcomes	NOUN	NNS	dobj
,	PUNCT	,	punct
and	CCONJ	CC	cc
personalizing	VERB	VBG	conj
treatment	NOUN	NN	compound
plans	NOUN	NNS	dobj
.	PUNCT	.	punct
For	ADP	IN	prep
example	NOUN	NN	pobj
,	PUNCT	,	punct
machine	NOUN	NN	compound
learning	NOUN	NN	compound
models	NOUN	NNS	nsubj
can	AUX	MD	aux
analyze	VERB	VB	ROOT
medical	ADJ	JJ	amod
images	NOUN	NNS	dobj
such	ADJ	JJ	amod
as	ADP	IN	prep
X	NOUN	NNS	pobj
-	NOUN	NNS	pobj
rays	NOUN	NNS	pobj
and	CCONJ	CC	cc
MRIs	PROPN	NNPS	conj
with	ADP	IN	prep
high	ADJ	JJ	amod
accuracy	NOUN	NN	pobj
,	PUNCT	,	punct
often	ADV	RB	advmod
matching	VERB	VBG	advcl
or	CCONJ	CC	cc
exceeding	VERB	VBG	conj
human	ADJ	JJ	amod
performance	NOUN	NN	dobj
.	PUNCT	.	punct
AI	PROPN	NNP	nsubj
also	ADV	RB	advmod
plays	VERB	VBZ	ROOT
a	DET	DT	det
crucial	ADJ	JJ	amod
role	NOUN	NN	dobj
in	ADP	IN	prep
drug	NOUN	NN	compound
discovery	NOUN	NN	pobj
by	ADP	IN	prep
analyzing	VERB	VBG	pcomp
chemical	ADJ	JJ	compound
compounds	NOUN	NNS	dobj
and	CCONJ	CC	cc
predicting	VERB	VBG	conj
their	PRON	PRP\$	poss
effectiveness	NOUN	NN	dobj
,	PUNCT	,	punct
reducing	VERB	VBG	advcl

both	DET	DT	det
time	NOUN	NN	dobj
and	CCONJ	CC	cc
cost	NOUN	NN	conj
.	PUNCT	.	punct
Despite	SCONJ	IN	prep
these	DET	DT	det
benefits	NOUN	NNS	pobj
,	PUNCT	,	punct
concerns	NOUN	NNS	nsubj
remain	VERB	VBP	ROOT
regarding	VERB	VBG	prep
data	NOUN	NNS	compound
privacy	NOUN	NN	pobj
,	PUNCT	,	punct
bias	NOUN	NN	conj
in	ADP	IN	prep
medical	ADJ	JJ	amod
datasets	NOUN	NNS	pobj
,	PUNCT	,	punct
and	CCONJ	CC	cc
the	DET	DT	det
need	NOUN	NN	conj
for	ADP	IN	prep
human	ADJ	JJ	amod
oversight	NOUN	NN	pobj
in	ADP	IN	prep
critical	ADJ	JJ	amod
healthcare	NOUN	NN	compound
decisions	NOUN	NNS	pobj
.	PUNCT	.	punct
	SPACE	_SP	dep

Column	Meaning
token.text	Actual word
pos_	Universal POS tag
tag_	Detailed POS tag
dep_	Dependency relation

POS Tag Frequency Analysis

```
from collections import Counter
```

```
pos_counts = Counter([token.pos_ for token in doc])
```

```
print(pos_counts)
```

```
Counter({'NOUN': 78, 'PUNCT': 43, 'VERB': 41, 'ADP': 32, 'ADJ': 30, 'PROPN': 16, 'CC  
ONJ': 14, 'AUX': 12, 'ADV': 12, 'DET': 11, 'PART': 5, 'PRON': 5, 'NUM': 3, 'SCONJ':  
3, 'SPACE': 3})
```

This helps analyze:

- Noun density
- Verb usage
- Sentence complexity

Dependency Parsing (Syntactic Structure)

Sentence-wise Dependency Analysis

```
for sent in doc.sents:  
    print(f"\nSentence: {sent}")  
    for token in sent:  
        print(f"{token.text:<12} --> {token.head.text:<12} ({token.dep_})")
```

Sentence: The Impact of Artificial Intelligence on Modern Society is Artificial Intelligence (AI) has emerged as one of the most transformative technologies of the twenty-first century.

The	-->	Impact	(det)
Impact	-->	is	(nsubj)
of	-->	Impact	(prep)
Artificial	-->	Intelligence	(compound)
Intelligence	-->	of	(pobj)
on	-->	Impact	(prep)
Modern	-->	Society	(compound)
Society	-->	on	(pobj)
is	-->	is	(ROOT)
Artificial	-->	Intelligence	(compound)
Intelligence	-->	is	(attr)
(	-->	Intelligence	(punct)
AI	-->	Intelligence	(appos)
)	-->	Intelligence	(punct)
has	-->	emerged	(aux)
emerged	-->	is	(conj)
as	-->	emerged	(prep)
one	-->	as	(pobj)
of	-->	one	(prep)
the	-->	technologies	(det)
most	-->	transformative	(advmod)
transformative	-->	technologies	(amod)
technologies	-->	of	(pobj)
of	-->	technologies	(prep)
the	-->	century	(det)
twenty	-->	first	(advmod)
-	-->	first	(punct)
first	-->	century	(amod)
century	-->	of	(pobj)
.	-->	is	(punct)

Sentence: From simple rule-based systems to complex deep learning architectures, AI has evolved rapidly and is now deeply integrated into many aspects of daily life.

From	-->	evolved	(prep)
simple	-->	systems	(amod)
rule	-->	based	(npadvmod)
-	-->	based	(punct)
based	-->	systems	(amod)
systems	-->	From	(pobj)
to	-->	From	(prep)
complex	-->	learning	(amod)
deep	-->	learning	(amod)
learning	-->	to	(pobj)
architectures	-->	From	(pobj)
,	-->	evolved	(punct)
AI	-->	evolved	(nsubj)
has	-->	evolved	(aux)
evolved	-->	evolved	(ROOT)
rapidly	-->	evolved	(advmod)
and	-->	evolved	(cc)
is	-->	integrated	(auxpass)
now	-->	integrated	(advmod)
deeply	-->	integrated	(advmod)

integrated	--> evolved	(conj)
into	--> integrated	(prep)
many	--> aspects	(amod)
aspects	--> into	(pobj)
of	--> aspects	(prep)
daily	--> life	(amod)
life	--> of	(pobj)
.	--> evolved	(punct)

Sentence: Its influence can be seen across industries such as healthcare, education, finance, transportation, and entertainment.

Its	--> influence	(poss)
influence	--> seen	(nsubjpass)
can	--> seen	(aux)
be	--> seen	(auxpass)
seen	--> seen	(ROOT)
across	--> seen	(prep)
industries	--> across	(pobj)
such	--> as	(amod)
as	--> industries	(prep)
healthcare	--> as	(pobj)
,	--> healthcare	(punct)
education	--> healthcare	(conj)
,	--> education	(punct)
finance	--> education	(conj)
,	--> finance	(punct)
transportation	--> finance	(conj)
,	--> transportation	(punct)
and	--> transportation	(cc)
entertainment	--> transportation	(conj)
.	--> seen	(punct)

Sentence: As AI systems continue to grow in capability and adoption, they are reshaping how humans work, communicate, and make decisions.

As	--> continue	(mark)
AI	--> systems	(compound)
systems	--> continue	(nsubj)
continue	--> reshaping	(advcl)
to	--> grow	(aux)
grow	--> continue	(xcomp)
in	--> grow	(prep)
capability	--> in	(pobj)
and	--> capability	(cc)
adoption	--> capability	(conj)
,	--> reshaping	(punct)
they	--> reshaping	(nsubj)
are	--> reshaping	(aux)
reshaping	--> reshaping	(ROOT)
how	--> work	(advmod)
humans	--> work	(nsubj)
work	--> reshaping	(ccomp)
,	--> work	(punct)
communicate	--> work	(conj)
,	--> communicate	(punct)

```

and          --> communicate (cc)
make         --> communicate (conj)
decisions    --> make         (dobj)
.            --> reshaping   (punct)

--> .        (dep)

```

Sentence: At its core, artificial intelligence refers to the ability of machines to simulate human intelligence.

```

At           --> refers       (prep)
its          --> core         (poss)
core         --> At           (pobj)
,            --> refers       (punct)
artificial   --> intelligence (amod)
intelligence --> refers       (nsubj)
refers       --> refers       (ROOT)
to           --> refers       (prep)
the          --> ability      (det)
ability      --> to           (pobj)
of           --> ability      (prep)
machines     --> of           (pobj)
to           --> simulate     (aux)
simulate     --> ability      (acl)
human        --> intelligence (amod)
intelligence --> simulate     (dobj)
.            --> refers       (punct)

```

Sentence: This includes learning from data, recognizing patterns, understanding language, and making decisions with minimal human intervention.

```

This         --> includes     (nsubj)
includes     --> includes     (ROOT)
learning     --> includes     (xcomp)
from         --> learning     (prep)
data         --> from         (pobj)
,            --> learning     (punct)
recognizing  --> learning     (advcl)
patterns     --> recognizing   (dobj)
,            --> patterns     (punct)
understanding --> patterns     (acl)
language     --> understanding (dobj)
,            --> understanding (punct)
and          --> understanding (cc)
making       --> understanding (conj)
decisions    --> making       (dobj)
with         --> decisions    (prep)
minimal      --> intervention (amod)
human        --> intervention (amod)
intervention --> with         (pobj)
.            --> includes     (punct)

```

Sentence: Early AI systems were limited in scope and required explicit programming.

```

Early        --> systems      (amod)
AI           --> systems      (compound)
systems       --> limited     (nsubjpass)
were          --> limited     (auxpass)

```

limited	--> limited	(ROOT)
in	--> limited	(prep)
scope	--> in	(pobj)
and	--> limited	(cc)
required	--> limited	(conj)
explicit	--> programming	(amod)
programming	--> required	(dobj)
.	--> limited	(punct)

Sentence: However, modern AI relies heavily on machine learning and deep learning techniques, enabling systems to improve automatically through experience.

However	--> relies	(advmod)
,	--> relies	(punct)
modern	--> AI	(amod)
AI	--> relies	(nsubj)
relies	--> relies	(ROOT)
heavily	--> relies	(advmod)
on	--> relies	(prep)
machine	--> learning	(compound)
learning	--> on	(pobj)
and	--> learning	(cc)
deep	--> learning	(amod)
learning	--> techniques	(compound)
techniques	--> learning	(conj)
,	--> relies	(punct)
enabling	--> relies	(advcl)
systems	--> enabling	(dobj)
to	--> improve	(aux)
improve	--> enabling	(xcomp)
automatically	--> improve	(advmod)
through	--> improve	(prep)
experience	--> through	(pobj)
.	--> relies	(punct)

Sentence: The availability of large datasets, increased computational power, and advanced algorithms has accelerated this progress significantly.

The	--> availability	(det)
availability	--> increased	(nsubj)
of	--> availability	(prep)
large	--> datasets	(amod)
datasets	--> of	(pobj)
,	--> availability	(punct)
increased	--> increased	(ROOT)
computational	--> power	(amod)
power	--> increased	(dobj)
,	--> increased	(punct)
and	--> increased	(cc)
advanced	--> algorithms	(amod)
algorithms	--> accelerated	(nsubj)
has	--> accelerated	(aux)
accelerated	--> increased	(conj)
this	--> progress	(det)
progress	--> accelerated	(dobj)
significantly	--> accelerated	(advmod)

. --> accelerated (punct)

--> . (dep)

Sentence: One of the most impactful applications of AI is in healthcare.

One	--> is	(nsubj)
of	--> One	(prep)
the	--> applications	(det)
most	--> impactful	(advmod)
impactful	--> applications	(amod)
applications	--> of	(pobj)
of	--> applications	(prep)
AI	--> of	(pobj)
is	--> is	(ROOT)
in	--> is	(prep)
healthcare	--> in	(pobj)
.	--> is	(punct)

Sentence: AI-powered systems are now used to assist doctors in diagnosing diseases, predicting patient outcomes, and personalizing treatment plans.

AI	--> powered	(npadvmod)
-	--> powered	(punct)
powered	--> systems	(amod)
systems	--> used	(nsubjpass)
are	--> used	(auxpass)
now	--> used	(advmod)
used	--> used	(ROOT)
to	--> assist	(aux)
assist	--> used	(xcomp)
doctors	--> assist	(dobj)
in	--> assist	(prep)
diagnosing	--> in	(pcomp)
diseases	--> diagnosing	(dobj)
,	--> assist	(punct)
predicting	--> assist	(advcl)
patient	--> outcomes	(amod)
outcomes	--> predicting	(dobj)
,	--> predicting	(punct)
and	--> predicting	(cc)
personalizing	--> predicting	(conj)
treatment	--> plans	(compound)
plans	--> personalizing	(dobj)
.	--> used	(punct)

Sentence: For example, machine learning models can analyze medical images such as X-rays and MRIs with high accuracy, often matching or exceeding human performance.

For	--> analyze	(prep)
example	--> For	(pobj)
,	--> analyze	(punct)
machine	--> learning	(compound)
learning	--> models	(compound)
models	--> analyze	(nsubj)
can	--> analyze	(aux)
analyze	--> analyze	(ROOT)
medical	--> images	(amod)

images	--> analyze	(dobj)
such	--> as	(amod)
as	--> images	(prep)
X	--> as	(pobj)
-	--> as	(pobj)
rays	--> as	(pobj)
and	--> rays	(cc)
MRIs	--> rays	(conj)
with	--> analyze	(prep)
high	--> accuracy	(amod)
accuracy	--> with	(pobj)
,	--> analyze	(punct)
often	--> matching	(advmod)
matching	--> analyze	(advcl)
or	--> matching	(cc)
exceeding	--> matching	(conj)
human	--> performance	(amod)
performance	--> exceeding	(dobj)
.	--> analyze	(punct)

Sentence: AI also plays a crucial role in drug discovery by analyzing chemical compounds and predicting their effectiveness, reducing both time and cost.

AI	--> plays	(nsubj)
also	--> plays	(advmod)
plays	--> plays	(ROOT)
a	--> role	(det)
crucial	--> role	(amod)
role	--> plays	(dobj)
in	--> role	(prep)
drug	--> discovery	(compound)
discovery	--> in	(pobj)
by	--> plays	(prep)
analyzing	--> by	(pcomp)
chemical	--> compounds	(compound)
compounds	--> analyzing	(dobj)
and	--> analyzing	(cc)
predicting	--> analyzing	(conj)
their	--> effectiveness	(poss)
effectiveness	--> predicting	(dobj)
,	--> predicting	(punct)
reducing	--> predicting	(advcl)
both	--> time	(det)
time	--> reducing	(dobj)
and	--> time	(cc)
cost	--> time	(conj)
.	--> plays	(punct)

Sentence: Despite these benefits, concerns remain regarding data privacy, bias in medical datasets, and the need for human oversight in critical healthcare decisions.

Despite	--> remain	(prep)
these	--> benefits	(det)
benefits	--> Despite	(pobj)
,	--> remain	(punct)
concerns	--> remain	(nsubj)
remain	--> remain	(ROOT)

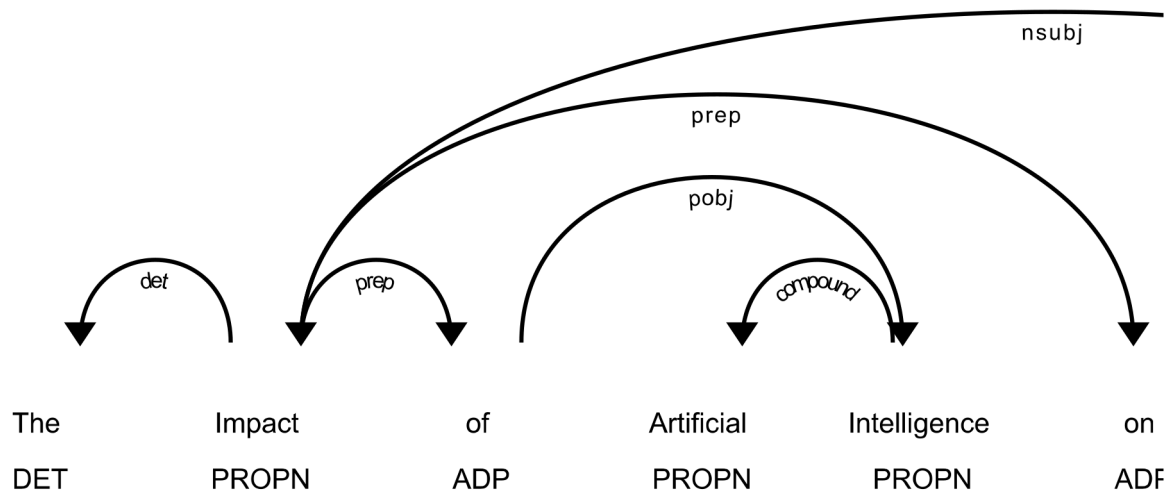
regarding	--> remain	(prep)
data	--> privacy	(compound)
privacy	--> regarding	(pobj)
,	--> privacy	(punct)
bias	--> privacy	(conj)
in	--> bias	(prep)
medical	--> datasets	(amod)
datasets	--> in	(pobj)
,	--> bias	(punct)
and	--> bias	(cc)
the	--> need	(det)
need	--> bias	(conj)
for	--> need	(prep)
human	--> oversight	(amod)
oversight	--> for	(pobj)
in	--> oversight	(prep)
critical	--> decisions	(amod)
healthcare	--> decisions	(compound)
decisions	--> in	(pobj)
.	--> remain	(punct)
	--> .	(dep)

In this:

- Head word of each token
- Grammatical relationship
- Sentence structure flow

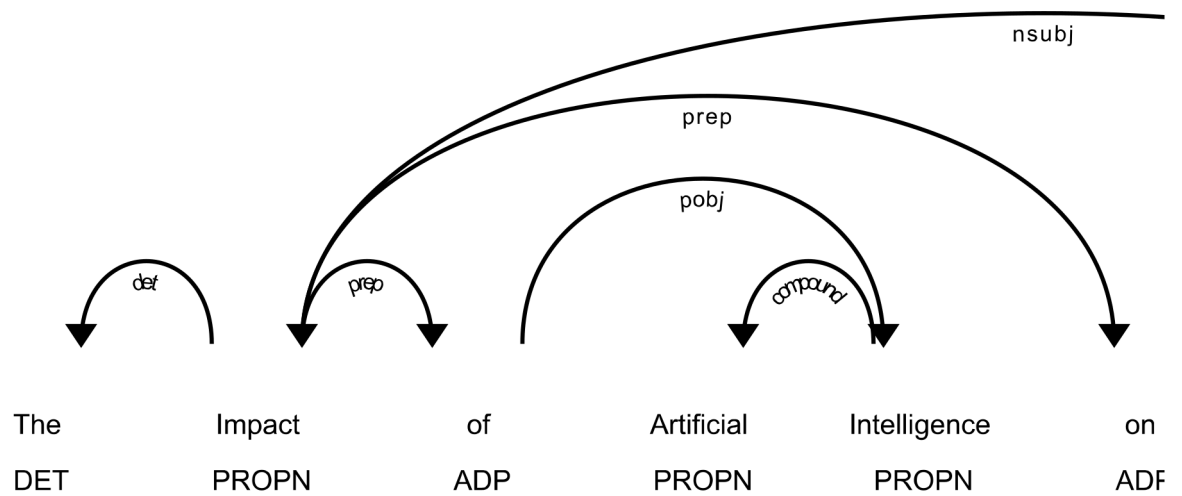
Single Sentence Dependency Tree Visualization

```
In [43]: sentence = list(doc.sents)[0]
displacy.render(sentence, style="dep", jupyter=True, options={'distance': 110})
```



All Sentence Dependency Tree Visualization

```
In [44]: from spacy import displacy  
  
displacy.render(doc, style="dep", jupyter=True, options={'distance': 110})
```



Key Components in the Visualization

Tokens (Words)

- Displayed horizontally in the order they appear in the sentence.
- Each token represents a single word.

ROOT

- The main verb of the sentence.
- Acts as the central node.
- All other words are directly or indirectly connected to the ROOT.

Example:

transforming → ROOT

options={'distance': 110})

Results

- spaCy accurately identified POS tags and syntactic relations
- Dependency tree visualization clearly represented sentence grammar
- spaCy performed faster and better than NLTK on large text

Conclusion

In this project, POS tagging and dependency parsing were successfully applied to a text corpus to analyze its grammatical structure. POS tagging provided word-level grammatical information, while dependency parsing offered deeper insights into syntactic relationships such as subject, object, and modifiers. The dependency tree visualization helped clearly illustrate these relationships in a structured and interpretable manner. Overall, this approach enhances the understanding of sentence structure and forms a strong foundation for advanced NLP tasks such as information extraction, question answering, and machine translation.